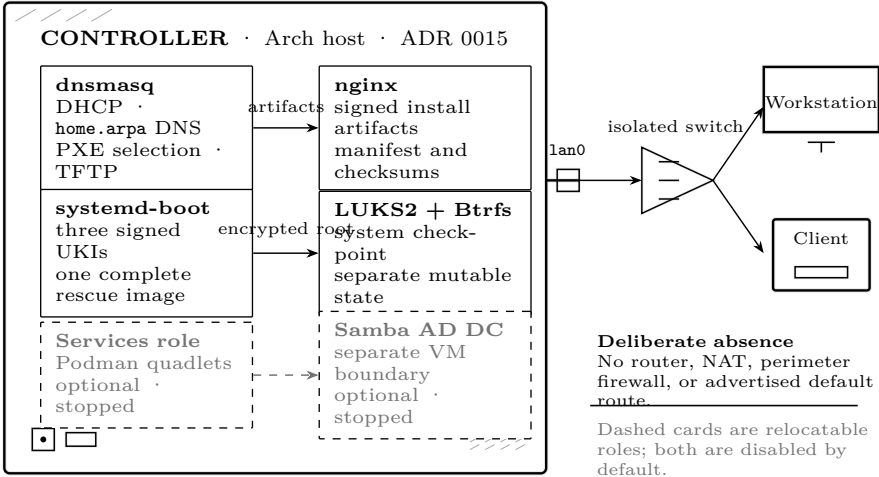


The Controller is the homelab’s infrastructure host. It answers DHCP and DNS for its managed network, serves the boot chain that installs every other machine, and is the one system that must be reconstructible without any other system being available. Everything else in the lab may depend on the Controller. The Controller may depend on nothing except a copy of this document, an installer image and a recorded set of inputs.

Scope

It owns	Boot, storage and encryption; the managed network interface; bundled DHCPv4 and <code>home.arpa</code> DNS; PXE selection and first-stage TFTP; the HTTP artifact service; local recovery; and its own checkpoint and rollback.
It may host	The optional Services role — Homebridge, openHAB and similar — and the identity domain controller in a VM. Both are separable and neither is required for the Controller to do its job. GOVERNED BY ADR 0054, ADR 0055
It does not do	Routing, NAT or perimeter firewalling. It has one network interface and advertises no default route. On the isolated acceptance network there is deliberately no path to the Internet. GOVERNED BY ADR 0011
It is not	A hypervisor-first platform. Foundational services run directly on the Arch host; containers and VMs are used where they earn their place, not by default. GOVERNED BY ADR 0014
One failure domain	Containers and VMs on the Controller share its hardware, its kernel maintenance window and its power supply. Placing a service on the Controller does not make it independently available.

Roles and boundaries



Storage

Part	Size	Why
ESP	2 GiB, FAT32	systemd-boot plus three UKIs, one of which carries a complete rescue userspace. Sized with deliberate headroom so an update can stage a full new artifact set before switching. The ESP is the one partition that cannot be grown afterwards without moving everything behind it. GOVERNED BY ADR 0051
Root	remainder, LUKS2	Btrfs inside the encrypted container. The operating system is one checkpointable subvolume; logs, caches, home directories, the snapshot store and mutable service data are separate subvolumes excluded from it. GOVERNED BY ADR 0026, ADR 0027
Swap	file in root	A file inside the encrypted Btrfs, so encrypted swap costs no second LUKS container. Hibernation is disabled — a machine whose job is answering DHCP does not suspend.

There is no Windows partition. ADR 0047 removed it: firmware maintenance uses **fwupd** where the vendor publishes to LVFS and vendor-supplied bootable media otherwise, which costs nothing until the day it is needed and removes a licence, a BitLocker recovery-key custody chain and a second Secure Boot trust path from the design.

The network boundary

One interface	Selected explicitly at install time and pinned by permanent MAC address to the stable name lan0 . There is no default and no automatic choice, because binding DHCP to the wrong link puts a second DHCP authority on a network the Controller does not own. GOVERNED BY ADR 0050
One authority	dnsmasq is the only address-assigning DHCP server on its layer-2 network. Where external infrastructure owns DHCP, the Controller's instance stays stopped. Two authorities on one segment is forbidden at every stage, including transitions. GOVERNED BY ADR 0008, ADR 0044
Bundled	DHCP and DNS start and stop as one unit. A partial success is a failure. GOVERNED BY ADR 0012
IPv4 only	Managed addressing is IPv4. The host's IPv6 stack stays enabled and unmanaged; nothing globally disables it. GOVERNED BY ADR 0013
Four inputs	The operator supplies subnet, Controller address, and pool start and end. Netmask, network and broadcast addresses, the DNS server, the suffix and the absence of a router are all derived. Contradictory values cannot reach dnsmasq because they cannot be entered. GOVERNED BY ADR 0045
Fail closed	First boot verifies the interface, its MAC, carrier, the address and the absence of a competing DHCP authority before starting anything. If any check fails, DHCP stays stopped and the machine reports why. GOVERNED BY ADR 0009

Lifecycle: install once, converge continuously

The single most important structural decision is that installation and configuration are different things with different cadences.

Stage	What happens
Install	Only what cannot be done later: partition, encrypt, make filesystems, install the base system and both kernels, write boot artifacts, configure lan0 and the static address, install sshd with the administrator public key, record the manifest. Then power off. GOVERNED BY ADR 0010, ADR 0053
Relocate	The operator physically moves the powered-off Controller to the isolated switch. Being powered off during the move is the primary DHCP-conflict safety boundary — not a configuration flag. GOVERNED BY ADR 0010
First boot	One-shot activation validates the network plan, then starts dnsmasq and nginx together, or leaves them stopped and says why.
Converge	Everything else is Ansible over SSH from this repository: package sets, service configuration, the Services role, identity client, users, sudo, health checks. Idempotent, reviewable as a diff, and the same mechanism for the Controller and every Workstation. GOVERNED BY ADR 0053
Update	Guarded: read the news, checkpoint, full-system upgrade, reboot if foundational, validate DHCP, DNS and PXE automatically, roll back if validation fails. GOVERNED BY ADR 0016, ADR 0028

Why this split matters

The previous design specified provisioning in detail and day-two configuration not at all, which meant that adding a service or changing a DHCP reservation implied reinstalling the machine. Nothing iterates at that cost, so in practice the running system drifts from the record and the record stops being true. The install stage is deliberately small so that it can be fully tested; everything that changes often lives where changes are cheap and reversible.

Optional roles

Services	Homebridge, openHAB and similar, as Podman quadlets with host networking — required, because HomeKit and UPnP discovery depend on mDNS and do not survive bridged container networking. USB radios bound by stable <code>by-id</code> path. Application state on its own subvolume with its own backup, because its restore point is not the operating system's. Disabled by default. GOVERNED BY ADR 0054
The trade-off	Running Services on the Controller means a Controller maintenance window is also a home-automation outage. That is accepted and recorded, not hidden. When it stops being acceptable, the role moves to its own host and nothing else changes.
Identity	Samba AD DC in a VM — a genuine separate trust boundary with its own restart and upgrade cadence, which is precisely when ADR 0014 permits a VM. Arch clients use SSSD's AD provider; Windows Pro clients domain-join; an AD-capable NAS can share the same identities. GOVERNED BY ADR 0055
Break-glass	Every managed machine keeps a separately named local administrator that is never a directory account. While the directory is down, only cached and break-glass logins work — so this is mandatory, not a nicety.

Milestones

A: functional proof	Network boot, interactive authorization, install, isolated-network activation, DHCP and DNS validation, checkpoint and rollback, Workstation pivot. Encryption is real throughout. Secure Boot may be disabled under an explicit, recorded development-proof mode with manual LUKS2 unlock at every boot, no production data, and no custom signing material anywhere. GOVERNED BY ADR 0043
B: hardening	Revisit ADRs 0029–0042 as a group with operational experience, then enforce Secure Boot, implement signing and custody, enable TPM unlock only after its recovery path is tested, and prove an unattended cold restart restores service. Reprovision from the proven installer; the proof installation is disposable by construction. GOVERNED BY ADR 0043
Known defects to fix in B	The key separation in ADRs 0033–0035 is notional once ADR 0039 gives both media one passphrase; the backup's independence is defeated by sharing that passphrase; and ADR 0030's additive trust model may be infeasible on firmware that only permits <code>db</code> changes from Setup Mode. All three are recorded rather than rediscovered. GOVERNED BY ADR 0057
No system passes	Final acceptance until Milestone B is complete.

THE BOOTSTRAP RULE

The Controller rebuild procedure must never require a functioning Controller to obtain the instructions, DNS records, installer image, artifact manifest or decryption material needed to rebuild it. Every recovery path in this design is tested with the Controller assumed dead. That is the whole reason this document is printable.

Telos. Homelab — Controller Design. Revised 2026-07-25. Project-created text and diagrams are licensed CC BY 4.0. Incorporated public-domain artwork remains public domain and is credited on the plate it appears with. See **LICENSE** and **CREDITS.md** in the source. Nothing here is professional advice; verify current regulations and follow the stated safety procedure.